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IMAGING SYSTEM WITH COMPOSITE GEARING AND METHOD THEREOF

Abstract

An x-ray imaging system is configured for mammography and tomosynthesis imaging modes. A rotational drive system is configured to support and drive rotational movement of an x-ray tube arm and a compression arm. A gearing system includes a driving gear and a driven gear. The driven gear is a composite gear with a tooth ring formed from a polymer material. The driving gear and the driven gear are in an over-meshed configuration such that a loading force is induced within the teeth of the driven gear engaged with the driving gear. The loading force reduces or prevents lash movement of the x-ray tube arm during rotation into and out of a 0° tube arm angle position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/559,889, filed Mar. 1, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Breast cancer and other breast lesions continue to be a significant threat to women's health. X-ray mammography and tomosynthesis are the most widely used tools for early detection, screening, and diagnosis. At least some known imaging systems that perform mammography and tomosynthesis imaging procedures have an upright gantry with an x-ray tube arm that at least partially rotates and a compression arm that at least partially rotates. A gear box with a gear system is typically used to facilitate the rotational movement of the x-ray tube arm and the compression arm. The gear system includes metal gears meshed with each other and with clearance between the gears so that lubrication can flow therebetween. As the x-ray tube arm and/or the compression arm rotates, the clearance between the gears can induce angular discontinuity jumps of the x-ray tube arm and/or compression arm because of positional variations in the supported weight. Accordingly, improvements to the rotational drive system of the x-ray tube arm and/or compression arm are desired.

SUMMARY

[0003] Examples of the disclosure are directed to composite gear systems for rotational drive assemblies in mammography and tomosynthesis imaging systems.

[0004] In an aspect, the technology relates to an x-ray imaging system including: a gantry including a rotational drive assembly; an x-ray tube arm including an x-ray source coupled to the gantry at the rotational drive assembly, the x-ray tube arm configured to selectively rotate relative to the gantry around a rotational axis via the rotational drive assembly; and an arm including an immobilization system coupled to the gantry and independently rotatable relative to the gantry around the rotational axis via the rotational axis, wherein the immobilization system includes a paddle, a support platform, and an x-ray receptor disposed below the support platform, wherein the rotational drive assembly includes: a tube shaft rotatable around the rotational axis and supporting the x-ray tube arm; an actuator configured to drive rotation of the tube shaft; and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, and wherein the driven gear is over-meshed with the driving gear such that the at least one second tooth is in direct contact with the at least one first tooth with the at least one second tooth at least partially deflected causing a loading force within the at least one second tooth.

[0005] In an example, the deflection of the at least one second tooth includes bending and compression of the at least one second tooth. In another example, the loading force is between 0.1 kilopound per square inch (ksi) and 5 ksi. In still another example, the loading force is measured when the x-ray tube arm is at a 0° tube arm angle. In yet another example, the loading force occurs when the x-ray tube arm is between at least a ±5° and a ±30° tube arm angles. In an example, the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.

[0006] In another example, the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring. In still another example, the rotational drive assembly includes

a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm. In yet another example, the driven gear is rotatable around the rotational axis and the driving gear is rotatable around a drive axis, and a distance between the drive axis and the rotational axis is such that the driving gear has an interference fit with the driven gear and a first pitch radius of the at least one first tooth at least partially overlaps with a second pitch radius of the at least one second tooth. In an example, the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth compresses the at least one second tooth inducing the loading force therein. In another example, the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth is between 1% and 5% of the distance between the drive axis and the rotational axis at a 0° tube arm angle in a fully-meshed configuration.

[0007] In still another example, the interference fit is measured when the x-ray tube arm is at a 0° tube arm angle. In yet another example, the interference fit occurs when the x-ray tube arm is between at least a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.

[0008] In another aspect, the technology relates to a method of rotating an x-ray tube arm of an imaging system, the imaging system including a gantry rotationally supporting the x-ray tube arm and an arm having an immobilization system including a paddle, a support platform, and an x-ray receptor disposed below the support platform, the method including: providing a rotational drive assembly, wherein the rotational drive assembly includes a tube shaft rotatable around a tube arm axis and supporting the x-ray tube arm, an actuator, and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, and wherein the driven gear is over-meshed with the driving gear such that a loading force is induced within the at least one second tooth; positioning, via the rotational drive assembly, the x-ray tube arm that includes an x-ray source at a 0° tube arm angle relative to the gantry; rotating the x-ray tube arm about the tube arm axis and out of the 0° tube arm angle by actuating the driving gear thereby rotating the driven gear around the tube arm axis; and based on the loading force within the at least one second tooth, preventing lash of the x-ray tube arm via the driven gear when a weight of the x-ray tube arm moves out of the 0° tube arm angle.

[0009] In an example, the method further includes inducing the loading force based on a force-displacement approach, a pressure approach, or a sensing approach. In another example, rotating the x-ray tube arm occurs during a tomosynthesis imaging mode of the imaging system. In still another example, the loading force prevents lash of the x-ray tube arm at least between a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.

[0010] In another aspect, the technology relates to an x-ray imaging system including: a gantry including a rotational drive assembly defining a rotational axis; an x-ray tube arm including an x-ray source coupled to the gantry at the rotational drive assembly, the x-ray tube arm configured to selectively rotate relative to the gantry via the rotational drive assembly; and an arm including an immobilization system coupled to the gantry at the rotational drive assembly and independently rotatable relative to the gantry, wherein the immobilization system includes a paddle, a support platform, and an x-ray receptor disposed below the support platform, wherein the rotational drive assembly includes: a tube shaft rotatable around the rotational axis and supporting the x-ray tube arm; an actuator configured to drive rotation of the tube shaft; and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft, rotatable around the rotational axis, and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, wherein the driving gear is rotatable around a drive axis, and wherein a distance between the drive axis and the rotational axis is such that the driving gear has an interference fit with the driven gear and a first pitch radius of the at least one first tooth at least partially overlaps with a second pitch radius of the at least one

second tooth.

[0011] In an example, the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one first tooth compresses the at least one second tooth inducing a loading force therein. In another example, the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth is between 1% and 5% of the distance between the drive axis and the rotational axis at a 0° tube arm angle in a fully-meshed configuration. In still another example, the interference fit is measured when the x-ray tube arm is at a 0° tube arm angle. In yet another example, the interference fit occurs when the x-ray tube arm is between at least a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles. In an example, the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.

[0012] In another example, the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring. In still another example, the rotational drive assembly includes a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] The following drawings are illustrative of particular examples of the present disclosure and therefore do not limit the scope of the present disclosure. The drawings are not to scale and are intended for use in conjunction with the explanations in the following detailed description.

[0014] Examples of the present disclosure will hereinafter be described in conjunction with the appended drawings, wherein like numerals denote like elements.

[0015] FIG. 1 is a schematic view of an exemplary imaging system.

[0016] FIG. 2 is a perspective view of the imaging system of FIG. 1.

[0017] FIG. 3 is a schematic view of a compression arm and a tube arm of the imaging system shown in FIGS. 1 and 2.

[0018] FIG. 4 is a front perspective view of a rotational drive assembly for rotation of the compression arm and the tube arm shown in FIG. 3.

[0019] FIG. 5 is a rear perspective view of the rotational drive assembly of FIG. 4.

[0020] FIG. 6 is a cross-sectional, perspective view of a tube shaft assembly of the rotational drive assembly shown in FIGS. 4-5.

[0021] FIG. 7 is a perspective view of gearsets of the drive assembly shown in FIGS. 4-5.

[0022] FIG. 8 is a cross-sectional schematic view of a tube arm gearset of the gearsets shown in FIG. 7.

[0023] FIG. 8A is an enlarged tooth detail of the tube arm gearset shown in FIG. 8.

[0024] FIG. 9 is a flowchart illustrating an exemplary method of rotating an x-ray tube arm of an imaging system.

DETAILED DESCRIPTION

[0025] Various examples will be described in detail with reference to the drawings, wherein like reference numerals represent like parts and assemblies throughout the several views. Reference to various examples does not limit the scope of the claims attached hereto. Additionally, any examples set forth in this specification are not intended to be limiting and merely set forth some of the many possible examples for the appended claims.

[0026] References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” “an example,” “an aspect,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily

include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other examples whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Moreover, one having skill in the art will understand the degree to which terms such as “about,” “approximately,” or “substantially” convey in light of the measurement techniques utilized herein. To the extent such terms may not be clearly defined or understood by one having skill in the art, the term “about” shall mean plus or minus ten percent.

[0027] Throughout this description, references to orientation (c.g., front(ward), rear(ward), top, bottom, back, right, left, upper, lower, etc.) of the components of the imaging system relate to their position when installed on and in use with the imaging system and are used for case of description and illustration only. No restriction is intended by use of the terms regardless of how the components of the imaging system are situated on their own. As used herein, the terms “axial” and “longitudinal” refer to directions and orientations, which extend substantially parallel to a centerline of the component or system. Moreover, the terms “radial” and “radially” refer to directions and orientations, which extend substantially perpendicular to the centerline of the component or system. In addition, as used herein, the term “circumferential” and “circumferentially” refer to directions and orientations, which extend arcuately about the centerline of the component or system.

[0028] X-ray imaging systems include an x-ray tube arm rotating around an axis for various imaging modes or procedures. In a more specific example, digital breast tomosynthesis (c.g., DBT) relies on the x-ray tube arm moving in a limited arc angle over a stationary target while sequential projection images are captured and then reconstructed to provide information about cancers obscured by normal tissue located above and below a lesion. The limited arm movement may include a sweep in the range of 11° to 60°. A stable movement of the x-ray tube arm without discontinuities in the motion profile are desired to facilitate clear and accurate projection images. Known drive systems for x-ray tube arms have an all-metal worm gearing system. The metal gears are set to have clearance between them for lubrication and assembly tolerances. This clearance induces backlash or lash movement of the x-ray tube arm during rotational movement. For example, when the x-ray tube arm moves out of a 0° tube arm angle position, the weight of the x-ray tube arm causes the x-ray tube arm to undesirably move though the clearance within the gears (e.g., lash travel) and may cause angular motion discontinuity (c.g., image artifacts) in the projection images. Wear also occurs within the metal gears, thereby further increasing lash distances and requiring periodic service for re-lubrication and re-adjustment.

[0029] As used herein, 0° tube arm angle position is in relation to the tube arm being at a 12 o'clock position on the gantry because the lash travel is relative to the position of the tube arm on the gantry. It is appreciated that when describing imaging positions of the tube arm a 0° angle may also be used that relates to the object being imaged or the detector and the position of the tube arm such that the x-ray beam intersects object being image or the detector at a substantially 90° angle. When describing imaging positions, the 0° angle and the corresponding position of the tube arm may vary depending on the type of view. For example, in an MLO view, the tube arm position starts at a 30°-60° angle relative to the gantry.

[0030] Worm gearing provides a number of benefits for x-ray imaging systems because it is robust, quiet, and at high numeric ratios, self-locking or resistant to back driving. Additionally, x-ray tube arm motion is typically low speed, high-torque, and reversible. In the examples described herein, an all metal worm gearing system is replaced with a composite gearing system so as to maintain the

benefits of the worm gearing system described herein. Additionally, the composite gearing system facilitates a reduction in noise during operation due to vibration damping, and additional efficiencies (e.g., torque transmission) due to lower friction and self-lubricating properties. A driving gear, the wormshaft, is a metal (e.g., steel) gear with a helical tooth form. The driven gear, the wormgear, is a composite gear with a polymer wormgear tooth ring secured to a steel core by a casting process. This composite gear system can operate without maintenance lubrication and achieves reduced tooth wear due to the reduced friction and the elastic properties of the polymer. In operation, a one-time break-in lubrication may be used for the composite gears during the initial operational period.

[0031] When merely replacing the all-metal gears with the composite gears and maintaining the typical tooth clearance spacing configuration of the gears, typically 0.002-0.004 inches, angular motion discontinuity can still occur in the imaging system. When the x-ray tube arm moves out of the 0° tube arm angle, the weight of the x-ray tube arm induces lash movement through the clearance spacing and image artifacts may occur. However, because of the material properties of the polymer teeth of the wormgear, additional gear tooth spacing configurations are enabled that can address the angular motion discontinuities induced by the inherent lash generated by the typical clearance spacing.

[0032] In operation, the loading on the gears is not uniform, and varies from essentially zero at the 0° tube arm angle position to a maximum load occurring at a $\pm 90^\circ$ tube arm angle position. While performing static overload testing with the composite wormgear set at the typical clearance spacing, it was observed that the angular deflections of the x-ray tube arm were completely reversible (e.g., elastic). These results are due to: (1) Wormgear tooth deflection from bending; (2) Wormgear tooth compression due to normal force between the metal wormshaft face and the contact patch on the plastic wormgear tooth; and (3) A shear gradient in the gear tooth ring acting as a windup mechanism between the gear tooth ring pitch circle and the metal core. Accordingly, by utilizing the elastic properties of the composite gear, the composite gear can be positioned relative to the wormshaft such that a loading force is induced within the wormgear teeth, and angular motion discontinuities of the tube arm are reduced or prevented when rotating out of the 0° tube arm angle position.

[0033] In examples, the loading force within the wormgear teeth can be predetermined for reducing or preventing lash movement of the x-ray tube arm at certain angular ranges, and for example, $\pm 3^\circ$, $\pm 7.5^\circ$, $\pm 15^\circ$, $\pm 30^\circ$, or more. This configuration reduces or prevents wear or additional stresses on the gears while maintaining reasonable lash movement of the x-ray tube arm at large angular ranges and that are more easily accounted for by load position encoders within the imaging system.

[0034] FIG. 1 is a schematic view of an exemplary imaging system **100**. FIG. 2 is a perspective view of the imaging system **100**. Referring concurrently to FIGS. 1 and 2, not every element described below is depicted in both figures. The imaging system **100** immobilizes a patient's breast **102** for x-ray imaging (either or both of mammography, tomosynthesis, or other imaging modalities) via a breast immobilizer unit **104** that includes a static breast support platform **106** and a moveable paddle **108**. Different paddles, each having different purposes, are known in the art. The breast support platform **106** and the paddle **108** each have a compression surface **110** and **112**, respectively, that move towards each other to compress, immobilize, stabilize, or otherwise hold and secure the breast **102** during imaging procedures. In known systems, the compression surface **110**, **112** is exposed so as to directly contact the breast **102**. Either or both of these compression surfaces **110**, **112** may be rigid plastic, a flexible plastic, a resilient foam, a mesh or screen, and so on. The support platform **106** also houses an image receptor **116** and, optionally, a detector rotating mechanism **118**, and optionally an anti-scatter grid (not depicted, but disposed above the image receptor **116**). The immobilizer unit **104** is in a path of an imaging beam **120** emanating from an x-ray source **122**, such that the beam **120** impinges on the image receptor **116**.

[0035] The immobilizer unit **104** is supported on a first support arm, also referred to as a

compression arm **124**. The paddle **108** is coupled to a paddle arm **134** that is configured to be raised and lowered along the compression arm **124** forming an immobilization area **160** for the breast **102**. The x-ray source **122** is supported on a second support arm, also referred to as a tube arm **126**. For mammography, the compression arm and tube arm **124**, **126** can rotate as a unit about an axis **128** between different imaging orientations such as CC and MLO, so that the imaging system **100** can take a mammogram projection image at each orientation. In operation, the image receptor **116** remains in place relative to the support platform **106** while an image is taken. The immobilizer unit **104** releases the breast **102** for movement of the compression arm and tube arm **124**, **126** to a different imaging orientation. For tomosynthesis, the compression arm **124** stays in place, with the breast **102** immobilized and remaining in place, while at least the tube arm **126** rotates the x-ray source **122** relative to the immobilizer unit **104** and the compressed breast **102** about the axis **128**. The imaging system **100** takes plural tomosynthesis projection images of the breast **102** at respective projection angles of the beam **120** relative to the breast **102**.

[0036] Concurrently and optionally, the image receptor **116** may be rotated relative to the breast support platform **106**, the axis parallel to the axis **128**, and in sync with the rotation of the tube arm **126**. The rotation can be a proportion of the angle of the rotation of the x-ray source **122** (c.g.,) $\pm 3.5^\circ$ but may also be through a different angle selected such that the beam **120** remains substantially in the same position on the image receptor **116** for each of the plural images. The rotation can be about an axis **130**, which can but need not be in the image plane of the image receptor **116**. The detector rotating mechanism **118** that is coupled to the image receptor **116** can drive the image receptor **116** in a rotating motion. For tomosynthesis imaging and/or CT imaging, the breast support platform **106** can be horizontal or can be at an angle to the horizontal, e.g., at an orientation similar to that for conventional MLO imaging in mammography. The imaging system **100** can be solely a mammography system, a CT system, or solely a tomosynthesis system, other modalities such as ultrasound, or a “combo” system that can perform multiple forms of imaging. An example of a system has been offered by the assignee hereof under the trade name 3Dimensions™ system.

[0037] When the system is operated, the image receptor **116** produces imaging information in response to illumination by the imaging beam **120** and supplies it to an image processor **132** for processing and generating breast x-ray images.

[0038] A system control and workstation unit **138** including software controls the operation of the system and interacts with the operator to receive commands and deliver information including processed-ray images. The system control and workstation unit **138** may include an automatic exposure control (AEC) engine that may be configured to calculate the AEC x-ray dosage needed to perform x-ray imaging, such as a tomosynthesis sweep, on a patient's breast.

[0039] The imaging system **100** includes a floor mount or base **140** for supporting the imaging system **100** on a floor. A gantry **142** extends upwards from the floor mount **140** and rotatably supports both the tube arm **126** and the compression arm **124**. The tube arm **126** and the compression arm **124** are configured to rotate discretely from each other and may also be raised and lowered along a face **144** of the gantry **142** so as to accommodate patients of different heights. The x-ray source **122** is disposed within the tube arm **126**. Together, the tube arm **126** and the compression arm **124** may be referred to as a C-arm **145**. In some examples, the compression arm **124** and/or the entire C-arm **145** may be configured to tilt relative to the face **144** (c.g., tilt upwards or downwards for patient positioning). In an example, the tilt angle may be between $\pm 10^\circ$ with outward tilt facilitating imaging seated persons and inward tilt facilitating imaging persons with stooped posture. The tilt axis is orthogonal to the axis **128** of the C-arm **145**.

[0040] A number of interfaces and display screens are disposed on the imaging system **100**. These include a foot display screen **146**, a gantry interface **148**, a support arm interface **150**, and a compression arm interface **152**. In general, the various interfaces **148**, **150**, and **152** may include one or more tactile buttons, knobs, switches, as well as one or more display screens, including

capacitive touch screens with graphic user interfaces (GUIs) so as to enable user interaction with and control of the imaging system **100**. In general, the foot display screen **146** is primarily a display screen, though a capacitive touch screen might be utilized if required or desired.

[0041] FIG. **3** is a schematic view of the compression arm **124** and the tube arm **126** of the imaging system **100** (shown in FIGS. **1** and **2**). The tube arm **126** includes the x-ray source **122** that is disposed a distance **162** from the rotational axis **128**. As described above, the x-ray source **122** is configured to rotate around the rotational axis **128** in order to facilitate imaging procedures. When the x-ray source **122** is positioned vertically, the tube arm **126** is at a 0° tube arm angle position. The x-ray source **122** can rotate in both a clockwise and counterclockwise direction relative to the 0° tube arm angle position, and in the example, towards a $\pm 90^\circ$ tube arm angle position. The x-ray source **122** can be positioned at any intermediate tube arm angle position $\pm A$ as required or desired. Movement of the tube arm **126** is driven by a rotational drive assembly **200** described further below in reference to FIGS. **4-6**. In examples, the x-ray source **122** can rotate past the $\pm 90^\circ$ tube arm angle position as required or desired.

[0042] The x-ray source **122** has a relatively large weight that the tube arm **126** is configured to support and transfer the supported load towards the gantry **142** (shown in FIG. **2**). Additionally, the system components are designed with a safety factor that further increases the loads that need to be supported. The weight of the tube arm **126** is off-center relative to the rotational axis **128**. When the x-ray source **122** is at the 0° tube arm angle position, a load path **164** for the weight of the tube arm **126** is substantially vertical and aligned with the rotational axis **128**. Accordingly, loading forces being transferred within the rotational drive assembly **200** are relatively low because there are few loads creating a moment arm relative to the rotational axis **128**. However, when the x-ray source **122** moves out of the 0° tube arm angle position, the tube arm **126** forms a moment arm force **166** relative to the rotational axis **128** with a load path that is horizontally offset $166h$ from the rotational axis **128**. The horizontal offset $166h$ induces moment arm loads within the rotational drive assembly **200** that the components within are configured to support. A maximum load **168** condition is when the x-ray source **122** is at the $\pm 90^\circ$ tube arm angle position as the load path is horizontally spaced the furthest from the rotational axis **128** and at the entire distance **162**. During movement of the tube arm **126**, the loading forces through the rotational drive assembly **200** are not uniform.

[0043] The compression arm **124** is also rotatable around the rotational axis **128**. The compression arm **124** includes the paddle **108**, the support platform **106**, and other components. As such, the compression arm **124** also induces loading forces through the rotational drive assembly **200** and similar to the forces described above because the weight of the compression arm **124** is not centered with the rotational axis **128**. In aspects, the loading forces induced by the compression arm **124** are smaller than those induced by the tube arm **126**.

[0044] Rotational movement of the x-ray source **122** and/or compression arm **124** are driven by the imaging mode being used. Mammography imaging modes can include a craniocaudal (CC) view that is performed with the x-ray source **122** at the 0° tube arm angle position, one or more mediolateral oblique (MLO) views that is performed with the x-ray source **122** between 30° and 60° on the gantry with the compression arm **124** positioned 90° relative to the x-ray source **122**, and other image types are also contemplated herein. Tomosynthesis imaging modes can include a sweep of the x-ray source **122** between $\pm 7.5^\circ$, wide angle tomosynthesis with a sweep of the x-ray source **122** up to and including $\pm 30^\circ$, and other angular positions are also contemplated herein. In examples, tomo sweeps can occur from the CC or the MLO 0° imaging angle starting position as required or desired. Other imaging modes are also contemplated herein such as stereotactic imaging, wide angle imaging, CT imaging, lateral imaging, etc. As described above, when the tube arm **126** is moved out of the 0° tube arm angle position, the configuration of its rotational support structure is known to result in an angular discontinuity jump due to the weight of the tube arm **126** and this jump can produce undesirable image artifacts.

[0045] FIG. 4 is a perspective view of a rotational drive assembly **200** for rotation of the compression arm **124** and the tube arm **126** (shown in FIG. 3). FIG. 5 is another perspective view of the rotational drive assembly **200**. Referring concurrently to FIGS. 4 and 5, the rotational drive assembly **200** is configured to be mounted at least partially within the gantry **142** (shown in FIG. 2) and drive rotational movement of the compression arm **124** and the tube arm **126** as described herein. The rotational drive assembly **200** includes a frame **202** for coupling the rotational drive assembly **200** within the gantry **142**. The frame **202** includes an eyebolt **204** for facilitating mounting the rotational drive assembly **200** within the gantry **142**.

[0046] A tube shaft assembly **206** is rotatably supported on the frame **202** at a hub **208**. The tube shaft assembly **206** includes a compression arm tube shaft **210** that is configured to support the compression arm **124** at a distal end and facilitate rotation thereof around the axis **128**. The tube shaft assembly **206** also includes a tube arm tube shaft **212** that is configured to support the tube arm **126** at a distal end and facilitate rotation thereof around the axis **128**. The tube shaft assembly **206** is described further below in reference to FIG. 6.

[0047] The rotational drive assembly **200** includes a compression arm actuator **214** configured to selectively drive rotation of the compression arm tube shaft **210** and a tube arm actuator **216** configured to selectively drive rotation of the tube arm tube shaft **212**. The actuators **214**, **216** are supported on the frame **202** and may be any type of actuator that facilitates driving rotational movement of the tube shaft assembly **206**. In the example, the actuators **214**, **216** are electrical motors that can drive rotation in two directions, a first rotational direction and an opposite second rotational direction.

[0048] A compression arm gearset **218** operationally couples the compression arm actuator **214** to the compression arm tube shaft **210** and a tube arm gearset **220** operationally coupled the tube arm actuator **216** to the tube arm tube shaft **212**. The compression arm gearset **218** includes a first driving gear **221** (shown in FIG. 7) coupled to a first drive shaft **222** that is rotatable around a first drive axis **224** and a first driven gear **226** that is coupled to a proximal end of the compression arm tube shaft **210**. In the example, the first driving gear **221** is supported in a first housing **228** coupled to the frame **202**. The tube arm gearset **220** similarly includes a second driving gear **230** (shown in FIG. 7) coupled a second drive shaft **232** that is rotatable around a second drive axis **234** and a second driven gear **236** that is coupled to a proximal end of the tube arm tube shaft **212**. The second driving gear **230** is supported in a second housing **238** coupled to the frame **202**. In the example, the first and second drive axes **224**, **234** are parallel to each other, and are orthogonal to the rotational axis **128** of the rotational drive assembly **200**. Brakes **240** are also engaged with each of the drive shafts **222**, **232**.

[0049] The rotational drive assembly **200** also includes a tilt arm plate **242** that facilitates the rotational drive assembly **200** tilting about pivot pins **244** on the frame **202** and relative to the face of the gantry **142** via a tilt angle encoder drive gear **246**. The actuation mechanisms for the tilting motion are not shown. One or more tilt limit switch actuation plates **248** are coupled to the tilt arm plate **242**. An analog encoder **250** is coupled to the frame **202** and engaged with the compression arm portion of the tube shaft assembly **206** via a timing belt **252**. The analog encoder **250** is configured to measure rotational position of the compression arm portion of the tube shaft assembly **206**. A compression arm rotation limit switch circuit board **254** prevents driving the compression arm rotation beyond its safe mechanical limits, and a cable guide **256** facilitates cable management through the rotational drive assembly **200** and communication with the compression arm **124** and the tube arm **126**.

[0050] FIG. 6 is a cross-sectional, perspective view of the tube shaft assembly **206** of the rotational drive assembly **200** (shown in FIGS. 4-5). The tube shaft assembly **206** includes the compression arm tube shaft **210** and the tube arm tube shaft **212**, the shafts **210**, **212** are independently rotatable around the rotational axis **128**. The compression arm tube shaft **210** has a distal end **258** configured to support the compression arm **124** (shown in FIGS. 1 and 2) and an opposite proximal end **260**

that the first driven gear **226** is coupled to. A pulley **262** and a cam **263** are disposed adjacent to the first driven gear **226** and couples the timing belt **252** to the encoder **250**, while the cam **263** actuates the limit switch on PCB **254** (shown in FIG. 5).

[0051] The tube arm tube shaft **212** is rotationally supported by a pair of first bearings **264** on the compression arm tube shaft **210** and positioned radially outside. A distal end **266** of the tube arm tube shaft **212** is configured to support the tube arm **126** (shown in FIGS. 1 and 2) and an opposite proximal end **268** that the second driven gear **236** is coupled to. In the example, the tube arm tube shaft **212** has an axial length that is shorter than the compression arm tube shaft **210**. The tube arm tube shaft **212** is rotationally supported by a pair of second bearings **270** within the hub **208** that statically mounts to the frame **202** of the rotational drive assembly **200**.

[0052] The compression arm tube shaft **210**, the first driven gear **226**, the tube arm tube shaft **212**, and the second driven gear **236** are all coaxial relative to the rotational axis **128**. In the example, the first driven gear **226** has a plurality of first teeth **272** (shown in FIG. 7) and the second driven gear **236** has a plurality of second teeth **274** (shown in FIG. 7). The teeth **272**, **274** are only illustrated schematically in FIG. 5. An outer diameter **276** of the first teeth **272** of the first driven gear **226** is smaller than an outer diameter **278** of the second teeth **274** of the second driven gear **236**. In an example, the first teeth **272** have a 70-tooth count with a 7 inch pitch diameter, and the second teeth **274** have an 80-tooth count with an 8 inch pitch diameter. In other examples, the outer diameter, tooth count, and pitch diameter may be different than the above listed configurations as required or desired.

[0053] FIG. 7 is a perspective view of the compression arm gearset **218** and the tube arm gearset **220** of the rotational drive assembly **200** (shown in FIGS. 4-5). In the example, the first driving gear **221** is meshed with the first driven gear **226** so that rotation of the first driving gear **221** drives rotation of the first driven gear **226**. The second driving gear **230** is meshed with the second driven gear **236** so that rotation of the second driving gear **230** drives rotation of the second driven gear **236**. In the example, the first driving gear **221** is a wormshaft with at least one first tooth **280** and the second driving gear **230** is a wormshaft with at least one second tooth **282**. The at least one first and second teeth **280**, **282** are helical teeth. The first and second driving gears **221**, **230** are formed from a metal material. In an aspect, the first driving gear **221** and the second driving gear **230** have identical tooth geometries.

[0054] In operation, the gearsets **218**, **220** facilitate rotational movement of the compression arm **124** and the tube arm **126** (both shown in FIG. 3) and via drive from the actuators. As described above, both the compression arm **124** and the tube arm **126** are heavy and can generate large off-center loads relative to the rotational axis **128** (shown in FIG. 6). For example, the compression arm **124** carries the x-ray detector components and the paddle drive components, and the tube arm **126** carries the x-ray source components that are disposed at a large radial distance from the rotational axis **128**. Additionally, while rotational speed of the compression arm **124** and the tube arm **126** need not be fast, accurate angular positioning is desired so as to facilitate the imaging procedures such as mammography, tomosynthesis, and the like. Accordingly, using the worm gearing described herein provides for a low-speed, high-torque drive system that enables rotation in both rotational directions and is robust, quiet, and self-locking and/or resistance to back driving. It is appreciated that the rotational drive assembly gearsets may use other types of gears as required or desired, and such as, pinion gears, ring and pinion gears, etc.

[0055] In the example, both the first and second driven gears **226**, **236** are composite gears with a cylindrical metal core **284** and a polymer outer layer **286** forming the respective teeth **272**, **274**. In an aspect, the polymer outer layer **286** is cast on to the metal core **284**, and for example, overmolded to a knurled steel core. In one example, the polymer outer layer **286** is nylon 12. It is appreciated that other polymer materials may alternatively be used for the composite gears that are described herein. The alternative polymer materials may have the same or similar material properties to that of nylon 12. In aspects, a radial thickness of the polymer outer layer **286** is greater

than a radial thickness of the metal core **284**. In other aspects, the radial thickness of the polymer outer layer **286** is at least twice, three times, four times, or more the radial thickness of the metal core **284**. The polymer thickness and/or width may be based on speed and/or load bearing capacities of the gearing system.

[0056] When compared to all metal gearing, use of polymer on the first and second driven gears **226**, **236** provide a number of benefits. Friction between gears is reduced, thereby increasing gearset efficiencies. Torque transmission is increased, while noise, vibration, and/or heat generation are reduced during rotational operation. The first and second driven gears **226**, **236** described herein are also operable without maintenance lubrication thereby simplifying the rotational drive assembly for assembly and service. Wear is reduced or measurably eliminated during the operational lifetime of the gearsets and testing has demonstrated measurable wear is prevented of a rotational lifetime of 27 years, three times the nominal lifetime of such systems. In some examples, a one-time break in lubrication is applied to the gearsets prior to initial operation, but no further lubrication need be applied during the operational lifetime of the gearsets.

[0057] FIG. **8** is a schematic view of the tube arm gearset **220** and FIG. **8A** is a detailed view of the gearset mesh. Referring to FIGS. **8** and **8A**, while the tube arm gearset **220** is described below, it is appreciated that the compression arm gearset **218** (shown in FIG. **7**) is similarly designed. The tube arm gearset **220** includes the driving gear **230** that is configured to rotate around the drive axis **234** and the driven gear **236** that is configured to rotate around the rotational axis **128** (e.g., in and out of the page). The driving gear **230** is a wormshaft with a helical tooth **282**. The tooth **282** has a tooth depth **288** defined from a distal end **290** to a hub **292** of the gear **230**. An outer diameter **293** of the driving gear **230** is measured to the distal end **290** of the tooth **282**, while an inner diameter **294** of the driving gear **230** is measured at the hub **292**. The tooth **282** also includes a tooth thickness, measured orthogonal to the depth **288** and that tapers inwardly towards the distal end **290**, and a tooth pitch, measured circumferentially between the distal ends. A pitch radius **296** is defined from the drive axis **234** to the location on the tooth **282** that engages with the driven gear **236** in a theoretical meshed condition. In the example, the pitch radius **296** is disposed between the hub **292** and the distal end **290** of the tooth **282**.

[0058] The driven gear **236** is a wormgear with a plurality of teeth **274**. The teeth **274** each have a tooth depth **298** defined from a distal end **300** to a hub **302** of the gear **236**. The outer diameter **278** (shown in FIG. **6**) is measured to the distal end **300** of the teeth **274**, while an inner diameter **304** of the driven gear **236** is measured at the hub **302**. The teeth **274** also includes a tooth thickness, measured orthogonal to the depth **298** and that tapers inwardly towards the distal end **290**, and a tooth pitch, measured circumferentially between the distal ends. A pitch radius **306** is defined from the rotational axis **128** to the location on the teeth **274** that engages with the driving gear **230** in a theoretical meshed condition. In the example, the pitch radius **306** is disposed between the hub **302** and the distal end **300** of the teeth **274**.

[0059] Theoretically, aligning the pitch radius **296** of the driving gear **230** and the pitch radius **306** of the driven gear **236** places the gears **230**, **236** in a fully meshed condition for operation. Fully-meshed is used herein to describe the pitch radius **296** touching the pitch radius **306** at a tangential line to both circles. In practice, however, the gears **230**, **236** are under-meshed such that clearance is provided due to manufacturing tolerances and lubrication. Under-meshed is used herein to describe the pitch radius **296** spaced apart from the pitch radius **306** thereby providing clearance to the gear teeth. Under-meshing the driving gear **230** and the driven gear **236** described herein, however, results in an angular position jump when the tube arm **126** (shown in FIG. **3**) rotates out of the 0° tube arm angle position due to the loading conditions. This angular position jump due to movement loading on the gearset **220** can be referred to as backlash or lash and that is the clearance between the gear teeth **274**, **282** as described further above.

[0060] Accordingly, in the example, the gears **230**, **236** are over-meshed such that at least some of the engaged teeth **274** of the driven gear **236** are forced in direct contact against the driving gear

230. In examples, this direct contact of the engaged teeth **274** against the driving gear **230** induces a loading force within the engaged teeth **274** that acts against the tooth **282** of the driving gear **230** and restricts or prevents an angular position jump when the tube arm **126** rotates out of the 0° tube arm angle position. The loading force within the engaged teeth **274** at least partially counteracts the weight of the tube arm **126** that is being transferred through the gearset **220** as described above and so that lash does not occur. Over-meshing the gearset **220** is enabled because of the driven gear **236** having teeth **274** formed out of polymer material as described above and with material properties that enable elastic loading of the teeth **274**. It should be appreciated that over-meshing has an upper limit, whereby the loading force within the engaged teeth **274** is large enough to bind the gears **230**, **236** and restrict or prevent rotational operation of the gearset **220** or load the polymer teeth into an inelastic (e.g., plastic) zone. As such, the over-meshed configuration needs generate a loading force that is large enough to prevent or reduce lash, while also small enough not to bind the gears **230**, **236** or inelastically load the polymer teeth.

[0061] In an example, an over-mesh configuration of the gears may be defined by the pitch radius **296** of the driving gear **230** overlapping **308** the pitch radius **306** of the driven gear **236** such that the gear teeth **274**, **282** have an interference fit. As such, a distance **310** between the drive axis **234** and the rotational axis **128** is less than the pitch radius **296** and the pitch radius **306** added together and adjusting the distance **310** at least partially determines the loading force within the driven gear **236**. By reducing the distance **310** between the drive axis **234** and the rotational axis **128**, the teeth **274** of the driven gear **236** at least partially compress and or bend against the tooth **282** of the driving gear **230** thereby inducing a loading force within the teeth **274**.

[0062] In an aspect, the overlap **308** of the pitch radii **296**, **306** is between 1% and 20% of the distance **310** between the drive axis **234** and the rotational axis **128** in a fully-meshed configuration. The overlap **308** of the pitch radii **296**, **306** may be between 1% and 10% of the distance **310** between the drive axis **234** and the rotational axis **128** in a fully-meshed configuration. The overlap **308** of the pitch radii **296**, **306** may be between 1% and 5% of the distance **310** between the drive axis **234** and the rotational axis **128** in a fully-meshed configuration.

[0063] In another example, an over-mesh configuration may be defined by the deflection of the teeth **274** of the driven gear **236** when engaged with the tooth **282** of the driving gear **230** in the interference fit. The deflection of the teeth **274** may include both compression and bending stresses induced in the teeth and adjusting the compression and bending stresses at least partially determines the loading force within the driven gear **236**. Compression and bending stresses can be set by the distance **310** between gears as described above. Additionally, or alternatively, the shape and/or size of the teeth **274** can be used to determine the compression and bending stresses induced within the teeth **274** of the driven gear **236**. It is appreciated, that the shape and/or size of the driving gear teeth may additionally or alternatively be used to form the compression and bending stresses.

[0064] In an aspect, the loading force of the teeth **274** of the driven gear **236** is between 0.1 kilopound per square inch (ksi) and 12 ksi. The loading force of the teeth **274** of the driven gear **236** may be between 0.1 ksi and 9 ksi. The loading force of the teeth **274** of the driven gear **236** may be between 0.1 ksi and 5 ksi. The loading force of the teeth **274** of the driven gear is set when the tube arm **126** is at a 0° tube arm angle position.

[0065] As described above, the gear loading through the gearset **220** is not uniform. Larger loads are transferred through the gearset **220** as the tube arm **126** rotates closer to the $\pm 90^\circ$ tube arm angle position. As such, in one example, over-meshing the gears **230**, **236** may be configured to reduce or prevent lash though the entire loading cycle of the gearset **220** as the tube arm **126** rotates toward the $\pm 90^\circ$ tube arm angle position. In this example, the maximum loading condition when the tube arm **126** is at the $\pm 90^\circ$ tube arm angle position is determined and the corresponding loading force is induced within the driven gear **236**, thereby inducing a positive loading force

within the gears that is greater than the load of the tube arm **126**.

[0066] In other examples, however, it may not be desirable to induce such a large loading force within the driven gear. For example, such a loading force may exceed the elastic properties of the polymer material, motor power may not be sufficient to overcome such a loading force, such a loading force may cause the gears to at least partially bind, or such a loading force may induce overlash (e.g., further movement of the tube arm) when the tube arm is closer to the unloaded 0° tube arm angle position. Additionally, increasing movement to bring the gears closer together results in compression of the composite gear in the region of contact on both sides of the wormshaft, leading to increasing contact areas on the adjacent wormgear teeth, and limiting the loading force induced therein. As such, in other examples, over-meshing of the gears is configured so that lash is reduced or prevented for tube arm angle positions that are less than $\pm 90^\circ$. In an aspect, lash in the gears is reduced or prevented for a $\pm 60^\circ$ tube arm angle position, a $\pm 30^\circ$ tube arm angle position, a $\pm 15^\circ$ tube arm angle position, a $\pm 10^\circ$ tube arm angle position, a $\pm 7.5^\circ$ tube arm angle position, a $\pm 5^\circ$ tube arm angle position, or a $\pm 3^\circ$ tube arm angle position.

[0067] By reducing the tube arm angles that lash is restricted or reduced, the outside heavy loading conditions do not apply heavy loading stresses on the gear teeth and lash at the outer positions that does occur can be more easily accounted for by load position encoders within the imaging system. In these examples, lash is reduced or prevented at a predefined subset of tube arm angle positions around the 0° position so as to reduce wear on the system and to increase movement performance, but at the other positions, lash may occur. The tube arm angles that lash is restricted or reduced may be referred to as a zero lash region. Once outside of the zero-lash region, the loading force within the composite gear is configured to allow lash movement of the tube arm, but this lash movement may be reduced as compared to what would otherwise be present.

[0068] Determining the loading force or pitch radius overlap for the required or desired zero lash region can be performed in different ways. In one example, a force-displacement approach can be utilized. The steel worm gear is fixed and a calculated torque load or an equivalent tangential force is applied to the composite gear to measure tooth displacement at the pitch circle. The applied load can be based on empirically derived data and interpolated values for loads and lash (e.g., displacement) at tube arm angles from 0 - 90° . Using this information, an initial lash setting at a given force can be selected.

[0069] In another example, a pressure-based approach can be utilized. Tooth contact area can be characterized as the tooth pressure builds from zero to a maximum (e.g., at the high angle lash region) and this enables assembly of the magnitude of tooth pressure under both loaded and zero lash configurations. Unit pressure scales fairly linearly from zero to a maximum, and as such, a relative large lash-setting tangential force can be applied so that the information can be used to set the pressure for the required or desired zero lash region. Tooth pressure may be measured, for example, by painting the wormshaft with a machinist's dye, engaging a meshing force, and observing the resulting mesh area on the wormgear teeth.

[0070] In still another example, a sensing-based approach can be utilized. The rotational movement of the wormgear is sensed as a desired tangential test force is reached. By measuring tooth displacement, an increasing tangential force is applied to the wormgear tooth. The tangential force is relatively small, and the information can be used to set the pre-load forces in the composite gear. An illustrative method to set zero-lash is to apply a criterion tangential force to any wormgear tooth, and adjust the pitch center distance such that tangential movement of that or another tooth is just observed with a sensitive instrument. For example, setting $\pm 7.5^\circ$ zero lash will load a gear tooth to 35 pounds force, while observing a maximum tangential tooth displacement of 0.0005 inches.

[0071] The foregoing discussions and setting methods pertain to the larger 80-tooth tube arm gear, with an 8.0-inch pitch diameter. The compression arm gear is smaller, with 70 teeth and a 7.0-inch pitch diameter. Since the worm gear teeth can be the same size and shape on both gears, the

tangential force to deflect them needs to be the same. Since the constant tangential force is applied at a smaller action radius, the equivalent torque the gear experiences is less, but not the tangential force. Additionally, composite gearing as described herein may be utilized in other drive components of the imaging system. For example, a paddle drive is disposed within the compression arm and is configured to move the paddle. The paddle drive typically includes a leadscrew and nut in combination with spur gearing driving linear movement of the paddle. The nut may be formed as a composite element so as to reduce lash within this type of system.

[0072] The determined loading of the composite gear for overmeshing can be implemented by either pushing the wormgear tooth with the corresponding tangential force or torquing the wormgear to achieve the same tangential force.

[0073] The over-mesh configuration may be measured and set in the field, or on an assembled machine, through reporting the angular position of the tube arm. For example, the angular position of the tube arm is measured and reported with a resolution to 0.01° . A spring scale capable of measuring up to a 5-pound force is provided. The tube arm and the compression arm are moved to the 0° angular position. Applying a 4.5-pound force in both directions on the tube arm and determining that the total movement is 0.01° or less. If setting is required, positional adjustment of the driven gear can be made. These steps are repeated for the compression arm with a 5-pound force and so that the total movement is 0.01° or less.

[0074] FIG. 9 is a flowchart illustrating an exemplary method **400** of rotating an x-ray tube arm of an imaging system. The example methods and operations can be implemented or performed by the assemblies described herein (e.g., the imaging system **100** and rotational drive assembly **200** shown in FIGS. 1-8). The imaging system includes a gantry rotationally supporting the x-ray tube arm and an arm having an immobilization system including a paddle, a support platform, and an x-ray receptor disposed below the support platform.

[0075] The method **400** begins with providing a rotational drive assembly (operation **402**). The rotational drive assembly includes a tube shaft rotatable around a tube arm axis and supporting the x-ray tube arm, an actuator, and a gearset. The gearset includes a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth. The at least one second tooth of the driven gear is formed from a polymer material, and the driven gear is over-meshed with the driving gear such that a loading force is induced within the at least one second tooth. The x-ray tube including an x-ray source is positioned at a 0° tube arm angle relative to the gantry via the rotational drive assembly (operation **404**).

[0076] The x-ray tube arm is rotated about the tube arm axis and out of the 0° tube arm angle by actuating the driving gear thereby rotating the driven gear around the tube arm axis (operation **406**). Based on the loading force within the at least one second tooth, lash of the x-ray tube arm assembly is prevented via the driven gear when a weight of the x-ray tube arm assembly moves out of the 0° tube arm angle (operation **408**). In examples, rotation of the x-ray tube arm is part of an imaging mode, such as tomosynthesis, so that image artifacts are reduced or eliminated in the corresponding images. The loading force may be generated based on a force-displacement approach, a pressure approach, a sensing approach (all described above), or any other setting that is developed in the future.

[0077] In examples, during rotation of the tube arm there is a continuous and seamless transition from the overmeshed gear teeth condition at zero/low/medium loading to the undermeshed gear teeth condition at higher loads. This transition can be calculated and adjusted to provide relatively uniform and consistent tooth unit pressures from zero to maximum tube arm and compression arm loadings (e.g., from 0° to) $\pm 90^\circ$. Empirical testing has validated these computed load pressures, and even when the gearset is highly overmeshed, safe gear tooth pressures are obtained over the full range of possible motion such as with the tube arm near the machine base at 180° (e.g., the transport position of mobile installations). This uniform tooth loading system has the benefit of reducing or eliminating motion discontinuity during loading from the 0° position. Additionally,

another benefit is making the drive motor loading more uniform, thereby facilitating driving the C-arm subsystems at a consistent speed. Accordingly, more uniform motion control, reduced motion artifacts, and/or reduced drivetrain noise are all facilitated and provide improvements to the known motion controllers that use nested control loops and algorithms.

EXAMPLES

[0078] Illustrative examples of the systems and methods described herein are provided below. An embodiment of the system or method described herein may include any one or more, and any combination of, the clauses described below:

[0079] Clause 1. An x-ray imaging system including: a gantry including a rotational drive assembly; an x-ray tube arm including an x-ray source coupled to the gantry at the rotational drive assembly, the x-ray tube arm configured to selectively rotate relative to the gantry around a rotational axis via the rotational drive assembly; and an arm including an immobilization system coupled to the gantry and independently rotatable relative to the gantry around the rotational axis via the rotational axis, wherein the immobilization system includes a paddle, a support platform, and an x-ray receptor disposed below the support platform, wherein the rotational drive assembly includes: a tube shaft rotatable around the rotational axis and supporting the x-ray tube arm; an actuator configured to drive rotation of the tube shaft; and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, and wherein the driven gear is over-meshed with the driving gear such that the at least one second tooth is in direct contact with the at least one first tooth with the at least one second tooth at least partially deflected causing a loading force within the at least one second tooth.

[0080] Clause 2. The x-ray imaging system of any one of clauses 1-28, wherein the deflection of the at least one second tooth includes bending and compression of the at least one second tooth.

[0081] Clause 3. The x-ray imaging system of any one of clauses 1-28, wherein the loading force is between 0.1 kilopound per square inch (ksi) and 5 ksi.

[0082] Clause 4. The x-ray imaging system of any one of clauses 1-28, wherein the loading force is measured when the x-ray tube arm is at a 0° tube arm angle.

[0083] Clause 5. The x-ray imaging system of any one of clauses 1-28, wherein the loading force occurs when the x-ray tube arm is between at least a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.

[0084] Clause 6. The x-ray imaging system of any one of clauses 1-28, wherein the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.

[0085] Clause 7. The x-ray imaging system of any one of clauses 1-28, wherein the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring.

[0086] Clause 8. The x-ray imaging system of any one of clauses 1-28, wherein the rotational drive assembly includes a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm.

[0087] Clause 9. The x-ray imaging system of any one of clauses 1-28, wherein the driven gear is rotatable around the rotational axis and the driving gear is rotatable around a drive axis, and wherein a distance between the drive axis and the rotational axis is such that the driving gear has an interference fit with the driven gear and a first pitch radius of the at least one first tooth at least partially overlaps with a second pitch radius of the at least one second tooth.

[0088] Clause 10. The x-ray imaging system of any one of clauses 1-28, wherein the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth compresses the at least one second tooth inducing the loading force therein.

[0089] Clause 11. The x-ray imaging system of any one of clauses 1-28, wherein the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second

tooth is between 1% and 5% of the distance between the drive axis and the rotational axis at a 0° tube arm angle in a fully-meshed configuration.

[0090] Clause 12. The x-ray imaging system of any one of clauses 1-28, wherein the interference fit is measured when the x-ray tube arm is at a 0° tube arm angle.

[0091] Clause 13. The x-ray imaging system of any one of clauses 1-28, wherein the interference fit occurs when the x-ray tube arm is between at least a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.

[0092] Clause 14. The x-ray imaging system of any one of clauses 1-28, wherein the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.

[0093] Clause 15. The x-ray imaging system of any one of clauses 1-28, wherein the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring.

[0094] Clause 16. The x-ray imaging system of any one of clauses 1-28, wherein the rotational drive assembly includes a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm.

[0095] Clause 17. A method of rotating an x-ray tube arm of an imaging system, the imaging system including a gantry rotationally supporting the x-ray tube arm and an arm having an immobilization system including a paddle, a support platform, and an x-ray receptor disposed below the support platform, the method including: providing a rotational drive assembly, wherein the rotational drive assembly includes a tube shaft rotatable around a tube arm axis and supporting the x-ray tube arm, an actuator, and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, and wherein the driven gear is over-meshed with the driving gear such that a loading force is induced within the at least one second tooth; positioning, via the rotational drive assembly, the x-ray tube arm that includes an x-ray source at a 0° tube arm angle relative to the gantry; rotating the x-ray tube arm about the tube arm axis and out of the 0° tube arm angle by actuating the driving gear thereby rotating the driven gear around the tube arm axis; and based on the loading force within the at least one second tooth, preventing lash of the x-ray tube arm via the driven gear when a weight of the x-ray tube arm moves out of the 0° tube arm angle.

[0096] Clause 18. The method of any one of clauses 1-28, further including inducing the loading force based on a force-displacement approach, a pressure approach, or a sensing approach.

[0097] Clause 19. The method of any one of clauses 1-28, wherein rotating the x-ray tube arm occurs during a tomosynthesis imaging mode of the imaging system.

[0098] Clause 20. The method of any one of clauses 1-28, wherein the loading force prevents lash of the x-ray tube arm at least between a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.

[0099] Clause 21. An x-ray imaging system including: a gantry including a rotational drive assembly defining a rotational axis; an x-ray tube arm including an x-ray source coupled to the gantry at the rotational drive assembly, the x-ray tube arm configured to selectively rotate relative to the gantry via the rotational drive assembly; and an arm including an immobilization system coupled to the gantry at the rotational drive assembly and independently rotatable relative to the gantry, wherein the immobilization system includes a paddle, a support platform, and an x-ray receptor disposed below the support platform, wherein the rotational drive assembly includes: a tube shaft rotatable around the rotational axis and supporting the x-ray tube arm; an actuator configured to drive rotation of the tube shaft; and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft, rotatable around the rotational axis, and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, wherein the driving gear is rotatable around a drive axis, and wherein a distance between the drive axis and the rotational axis is such that the driving gear has an interference fit with the driven gear and a first pitch radius of the at

least one first tooth at least partially overlaps with a second pitch radius of the at least one second tooth.

[0100] Clause 22. The x-ray imaging system of any one of clauses 1-28, wherein the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one first tooth compresses the at least one second tooth inducing a loading force therein.

[0101] Clause 23. The x-ray imaging system of any one of clauses 1-28, wherein the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth is between 1% and 5% of the distance between the drive axis and the rotational axis at a 0° tube arm angle in a fully-meshed configuration.

[0102] Clause 24. The x-ray imaging system of any one of clauses 1-28, wherein the interference fit is measured when the x-ray tube arm is at a 0° tube arm angle.

[0103] Clause 25. The x-ray imaging system of any one of clauses 1-28, wherein the interference fit occurs when the x-ray tube arm is between at least a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.

[0104] Clause 26. The x-ray imaging system of any one of clauses 1-28, wherein the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.

[0105] Clause 27. The x-ray imaging system of any one of clauses 1-28, wherein the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring.

[0106] Clause 28. The x-ray imaging system of any one of clauses 1-28, wherein the rotational drive assembly includes a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm.

[0107] This disclosure described some examples of the present technology with reference to the accompanying drawings, in which only some of the possible examples were shown. Other aspects can, however, be embodied in many different forms and should not be construed as limited to the examples set forth herein. Rather, these examples were provided so that this disclosure was thorough and complete and fully conveyed the scope of the possible examples to those skilled in the art.

[0108] Although specific examples were described herein, the scope of the technology is not limited to those specific examples. One skilled in the art will recognize other examples or improvements that are within the scope of the present technology. Therefore, the specific structure, acts, or media are disclosed only as illustrative examples. Examples according to the technology may also combine elements or components of those that are disclosed in general but not expressly exemplified in combination, unless otherwise stated herein. The scope of the technology is defined by the following claims and any equivalents therein.

Claims

1. An x-ray imaging system comprising: a gantry including a rotational drive assembly; an x-ray tube arm including an x-ray source coupled to the gantry at the rotational drive assembly, the x-ray tube arm configured to selectively rotate relative to the gantry around a rotational axis via the rotational drive assembly; and an arm including an immobilization system coupled to the gantry and independently rotatable relative to the gantry around the rotational axis via the rotational axis, wherein the immobilization system includes a paddle, a support platform, and an x-ray receptor disposed below the support platform, wherein the rotational drive assembly includes: a tube shaft rotatable around the rotational axis and supporting the x-ray tube arm; an actuator configured to drive rotation of the tube shaft; and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, and wherein the driven gear is over-meshed with the driving gear such that the at least one

- second tooth is in direct contact with the at least one first tooth with the at least one second tooth at least partially deflected causing a loading force within the at least one second tooth.
2. The x-ray imaging system of claim 1, wherein the deflection of the at least one second tooth includes bending and compression of the at least one second tooth.
 3. The x-ray imaging system of claim 1, wherein the loading force is between 0.1 kilopound per square inch (ksi) and 5 ksi.
 4. The x-ray imaging system of claim 1, wherein the loading force is measured when the x-ray tube arm is at a 0° tube arm angle.
 5. The x-ray imaging system of claim 1, wherein the loading force occurs when the x-ray tube arm is between at least a $\pm 5^{\circ}$ and a $\pm 30^{\circ}$ tube arm angles.
 6. The x-ray imaging system of claim 1, wherein the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.
 7. The x-ray imaging system of claim 1, wherein the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring.
 8. The x-ray imaging system of claim 1, wherein the rotational drive assembly includes a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm.
 9. The x-ray imaging system of claim 1, wherein the driven gear is rotatable around the rotational axis and the driving gear is rotatable around a drive axis, and wherein a distance between the drive axis and the rotational axis is such that the driving gear has an interference fit with the driven gear and a first pitch radius of the at least one first tooth at least partially overlaps with a second pitch radius of the at least one second tooth.
 10. The x-ray imaging system of claim 9, wherein the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth compresses the at least one second tooth inducing the loading force therein.
 11. The x-ray imaging system of claim 9, wherein the overlap of the first pitch radius of the at least one first tooth and the second pitch radius of the at least one second tooth is between 1% and 5% of the distance between the drive axis and the rotational axis at a 0° tube arm angle in a fully-meshed configuration.
 12. The x-ray imaging system of claim 9, wherein the interference fit is measured when the x-ray tube arm is at a 0° tube arm angle.
 13. The x-ray imaging system of claim 9, wherein the interference fit occurs when the x-ray tube arm is between at least a $\pm 5^{\circ}$ and a $\pm 30^{\circ}$ tube arm angles.
 14. The x-ray imaging system of claim 9, wherein the driving gear is formed from a metal material, the driving gear is a wormshaft and the at least one first tooth is a helical tooth.
 15. The x-ray imaging system of claim 9, wherein the driving gear includes a metal core, the polymer material cast on the metal core and forming a tooth ring.
 16. The x-ray imaging system of claim 9, wherein the rotational drive assembly includes a second tube shaft supporting the arm, a second actuator configured to drive rotation of the second tube shaft, and a second gearset, the second gearset having a smaller diameter driven gear than the driven gear of the tube shaft of the x-ray tube arm.
 17. A method of rotating an x-ray tube arm of an imaging system, the imaging system including a gantry rotationally supporting the x-ray tube arm and an arm having an immobilization system including a paddle, a support platform, and an x-ray receptor disposed below the support platform, the method comprising: providing a rotational drive assembly, wherein the rotational drive assembly includes a tube shaft rotatable around a tube arm axis and supporting the x-ray tube arm, an actuator, and a gearset including a driving gear coupled to the actuator and having at least one first tooth and a driven gear coupled to the tube shaft and having at least one second tooth, wherein the at least one second tooth of the driven gear is formed from a polymer material, and wherein the

driven gear is over-meshed with the driving gear such that a loading force is induced within the at least one second tooth; positioning, via the rotational drive assembly, the x-ray tube arm that includes an x-ray source at a 0° tube arm angle relative to the gantry; rotating the x-ray tube arm about the tube arm axis and out of the 0° tube arm angle by actuating the driving gear thereby rotating the driven gear around the tube arm axis; and based on the loading force within the at least one second tooth, preventing lash of the x-ray tube arm via the driven gear when a weight of the x-ray tube arm moves out of the 0° tube arm angle.

18. The method of claim 17, further comprising inducing the loading force based on a force-displacement approach, a pressure approach, or a sensing approach.

19. The method of claim 17, wherein rotating the x-ray tube arm occurs during a tomosynthesis imaging mode of the imaging system.

20. The method of claim 17, wherein the loading force prevents lash of the x-ray tube arm at least between a $\pm 5^\circ$ and a $\pm 30^\circ$ tube arm angles.
